Beyond Access: Explainable Machine Learning for Fintech Adoption and Behavioural Segmentation in Pakistan

Abdur-Rehman Ashar
Beaconhouse School System PTC
Gujranwala, Pakistan
abdurrehmanashar@gmail.com

Umema Ashar
KDD Lab, FAST NUCES
Islamabad, Pakistan
umema2004@gmail.com

***Abstract*—Financial inclusion policy in Pakistan, as in much of the developing world, is typically measured by formal account ownership — but ownership does not tell us whether people actually use digital financial tools once they have access. Using individual-level data from the Global Findex Database 2025 (Pakistan, $n = 1{,}000$), this study finds that 44.3% of the Findex Pakistan analytical sample belongs to a distinct segment we call "Connected but Unengaged": near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet two-thirds (66.8%) remain formally excluded from the financial system — nearly ten times the directly-measured dormant-account rate (6.6%). This segment is not an artifact of a single method: it emerges independently from unsupervised clustering, from the population profile of the classifier's prediction errors, and from a feature-group ablation study showing behavioural variables, not access variables, drive adoption in this context. The convergence of three unrelated analytical approaches on the same population boundary is the paper's central evidence that, for a large share of Pakistanis, this suggests that connectivity alone does not explain non-adoption and points toward downstream barriers – such as trust, product design, or institutional friction – that are not captured in survey data. We support this finding with an explainable ML pipeline (Random Forest, XGBoost, SHAP; macro F1 = 0.494, validated via repeated cross-validation) and a bootstrap-validated cluster structure (Adjusted Rand Index = 0.904). We also find that gender, not urban/rural status, is the strongest structural predictor of segment membership (Cramer's $V = 0.807$), and that naive subgroup accuracy comparisons are misleading once gender's entanglement with outcome is accounted for — a methodological caveat likely to generalize beyond this setting. All findings are associational; the cross-sectional nature of the data does not support causal claims about why this segment remains unengaged.**

***Index Terms*—fintech adoption, financial inclusion, explainable AI, SHAP, k-means clustering, model calibration, algorithmic fairness, Pakistan, Global Findex**

I. Introduction

A dormant-account rate of 6.6% captures inactivity among people who already hold an account, but does not capture the potentially larger population who have digital access yet never enter the formal financial system. This paper shows that number is almost certainly the wrong denominator. Using individual-level Findex 2025 data, we find that when access is measured properly — mobile phone ownership and internet use, rather than account status alone — a much larger population (44.3% of the analytical sample) has the technical capacity to engage digitally but does not. This gap is invisible to the standard dormant-account metric, which only counts people who already hold an account; it misses everyone who has connectivity but never obtained one in the first place. We surface this segment independently through three unrelated analytical methods and show the pattern holds under model-error inspection, feature-group ablation, and bootstrap validation. Establishing that this population exists and is measurable is the contribution of this paper; explaining why its members remain unengaged despite access is left as the natural next question.

This study applies explainable machine learning and unsupervised segmentation to individual-level survey data to identify who is excluded from Pakistan's digital financial system and, more specifically, where formal access fails to translate into actual digital usage. Using Random Forest and XGBoost classifiers with SHAP explainability alongside k-means clustering, the analysis tests three hypotheses concerning the access-usage gap, the urban-rural adoption divide, and the relative importance of demographic versus access-based predictors of adoption. Beyond the core predictive and segmentation results, we subject every major finding to an explicit robustness check: hyperparameter-tuned models are evaluated under repeated cross-validation rather than a single train-test split; the SHAP-based feature-importance ranking is cross-checked against a feature-group ablation study using an entirely different methodology; and the cluster structure is validated via bootstrap resampling and an algorithm-independent comparison against agglomerative clustering. To our knowledge, prior work has not combined these supervised, explainability, segmentation, calibration, and robustness analyses in this specific Pakistan fintech-adoption setting.

Although the analytical sample is modest in size, the underlying microdata are multidimensional survey data spanning demographic, digital-access, financial-behaviour, and resilience constructs. The contribution therefore lies not in computational scale alone, but in demonstrating a reproducible machine-learning framework for extracting heterogeneous behavioural patterns from population-level financial-inclusion data.

A. Novelty Statement

This work makes four specific contributions beyond prior literature.

First, it identifies and triangulates — via clustering, model-error analysis, and ablation — a large access-rich, adoption-poor population segment that is substantially larger than the directly-measured dormant-account rate would suggest, demonstrating the value of combining multiple analytical lenses rather than relying on a single metric.

Second, it provides an explicit robustness and calibration audit rarely reported in this literature, showing that headline accuracy figures can mask systematic underconfidence for the most policy-relevant class (Excluded) and that naive subgroup accuracy comparisons can be misleading when a demographic variable is strongly entangled with outcome.

Third, it offers a direct, quantitative test of whether the access-dominant predictor ranking reported for Sub-Saharan Africa [3] generalizes to Pakistan, finding that it does not — behavioural and demographic variables outrank raw access variables in this context, a result confirmed independently through SHAP, permutation importance, and ablation.

Fourth, it extends the dual supervised-unsupervised explainable ML methodology of Chowdhury et al. [1] to South Asian survey data, using the newer 2024-fieldwork Findex round rather than the 2021 round used in comparable Sub-Saharan African studies [3].

B. Research Questions

The analysis is organized around four research questions, which the hypotheses and analyses in Sections III–IV address individually and, in combination, jointly.

RQ1. To what extent does formal financial access translate into digital financial usage among Pakistani adults, and does this translation vary by urban/rural context?

RQ2. Can unsupervised learning, applied independently of any adoption label, identify distinct behavioural/access population profiles — and if so, do these profiles reveal structure not visible in directly-measured adoption statistics (e.g., the dormant-account rate)?

RQ3. Which demographic, access, behavioural, and resilience variables are most strongly associated with differences in Adoption Tier, and does the resulting ranking generalize across explanation methods (SHAP, Gini, permutation) and across national contexts (cf. Sub-Saharan Africa [3])?

RQ4. Are the population segments identified by clustering robust — to resampling, to the choice of clustering algorithm, and to corroboration through independent diagnostic routes (model error analysis, feature-group ablation)?

Table I maps each research question to the hypotheses tested in Section IV and the methods used to address it. RQ1 and RQ2 are addressed jointly: RQ1 through the directly-measured access-usage gap (H1) and the urban/rural comparison (H2), and RQ2 through the independent unsupervised segmentation that corroborates and extends the H1 finding (Section IV-A). RQ3 corresponds to H3, the supervised modelling and explainability analysis. RQ4 is not tested via a numbered hypothesis; instead, it is addressed directly through the cluster validation suite (Section IV-F) — bootstrap resampling, the Ward agglomerative cross-check, and the $k = 2$ vs. $k = 3$ comparison — and through the convergence of clustering, error analysis, and ablation summarized in Table VIII.

TABLE I
MAPPING OF RESEARCH QUESTIONS TO HYPOTHESES AND METHODS

RQ	Addressed by	Section(s)
RQ1	H1 (access-usage gap), H2 (urban/rural divide)	§IV-A, §IV-B
RQ2	Unsupervised k-means segmentation, independent of Adoption Tier	§III-L, §IV-A
RQ3	H3 (supervised modelling); SHAP, Gini, permutation importance	§III-E–§III-I, §IV-C
RQ4	Bootstrap ARI, Ward cross-check, $k = 2$ vs. $k = 3$ comparison, three-method convergence (Table VIII)	§IV-F, §V

II. RELATED WORK

A. Measuring Financial Inclusion

The Global Findex Database, now in its fifth edition, remains the primary demand-side data source for cross-country financial inclusion research [6]. Early editions of Findex research documented steady global growth in account ownership but repeatedly noted that account ownership alone is an incomplete proxy for financial inclusion, since it does not capture whether an account is actively used. Suri and Jack [5] provided some of the most influential causal evidence on this front, showing via a long-run panel study in Kenya that access to mobile money – specifically, active usage rather than mere availability – reduced extreme poverty and produced disproportionate gains for female-headed households through occupational shifts into business and retail activity. Their finding that gendered impacts of digital financial services can be large and structurally significant directly motivates our attention to gender as a candidate predictors of modelled fintech adoption, rather than treating it as a routine demographic control variable.

B. Machine Learning Approaches to Fintech Adoption

Three recent studies directly inform the design of this project. Chowdhury et al. [1] applied Random Forest, XGBoost, and k-means clustering to financial literacy and behaviour data from 1,067 adults in Bangladesh, validating their segmentation with silhouette, Davies-Bouldin, and Calinski-Harabasz indices. They identified three behavioural clusters (34%, 41%, and 25% of the sample) and reported a weak correlation between financial knowledge and financial behaviour, alongside a counterintuitive 7.3 percentage-point rural literacy advantage over urban respondents, with formal education emerging as a comparatively weak predictor relative to institutional trust.

Emuron, Kapim Kenfack, and Msimanga [2] took a more applied approach, predicting fintech adoption in South Africa from Findex, mobile pricing, and central bank data using regression-based methods on remittances, borrowing, age, and digital usage variables. Their work established a directly comparable variable set to the one constructed here from Findex Pakistan, though without a segmentation component or an explicit robustness audit.

Saidy and Hassan [3] modelled fintech adoption drivers across 36 Sub-Saharan African countries using the 2021 Global Findex round, combining Logit and Probit regression with Random Forest, Gradient Boosting, and XGBoost. Their central finding was that formal account ownership and technology access – specifically mobile phone ownership and internet connectivity – were the strongest predictors of adoption, a finding we directly test for generalizability in Section V.

C. Positioning

This project combines Chowdhury et al.'s dual supervised-unsupervised methodology with the adoption-prediction framing of Emuron et al. and Saidy and Hassan, applying both to Pakistan using the newer 2024-fieldwork Findex round [4] – a combination not yet applied to South Asian data. It further extends all three prior works methodologically by adding hyperparameter-tuned robustness estimates, SHAP dependence analysis, calibration diagnostics, cross-method feature importance validation, model error analysis linked back to the segmentation results, an ablation study, and bootstrap-validated cluster stability – none of which are reported in the three anchor papers.

III. Data and Methods

A. Data

The analysis uses the Global Findex Database 2025, Pakistan microdata (Reference: PAK_2024_FINDEX_v02_M) [4], an individual-level survey of 1,000 respondents conducted by the World Bank between May and December 2024. The dataset includes 183 variables spanning demographics, account ownership, digital and informal financial behaviour, and financial resilience. Findex microdata is collected under informed consent protocols administered by the World Bank and its survey partners and is distributed in de-identified form; no personally identifying information is available to or used by this analysis (see Section VI-E for further discussion). **All analyses in this paper use unweighted respondent counts; reported proportions describe the sample as collected and are not adjusted for Findex's survey design weights (Section VII). This choice reflects the study's focus on predictive modelling and behavioural segmentation within the observed analytical sample rather than estimation of population-level prevalence; consequently, reported proportions are interpreted as sample descriptors.**

B. Target Construction

A four-level ordinal Adoption Tier was constructed from account ownership and digital usage indicators: Excluded (no formal account), Basic (account held, no digital usage), Digitally Engaged (uses digital payments and/or a digitally enabled account), and Advanced (digital usage combined with formal saving or borrowing behaviour). The resulting distribution was 68.2% Excluded, 2.1% Basic, 10.1% Digitally Engaged, and 19.6% Advanced, shown in Fig. 2.

C. Feature Selection and Leakage Control

Twenty variables across four construct groups (demographics, access, behaviour, resilience) were initially selected from the Findex codebook as candidate predictors.

Leakage audit. An initial model run retaining all twenty candidate variables — including `account`, `dig_account`, `anydigpayment`, `saved`, and `fin22a` — produced spurious near-perfect classification accuracy. Inspection traced this to target leakage: the Adoption Tier label was constructed directly from these five variables (Section III-B), and `account_fin` and `account_mob` are near-deterministic components of `account` itself. In effect, the initial specification allowed the model to partially reconstruct its own target from near-identical inputs. **We report this failed initial specification deliberately, to demonstrate that leakage was detected and corrected before final evaluation**, rather than omitting it as if the leakage-safe specification had been the only one attempted.

All seven implicated variables were removed, leaving fifteen leakage-safe predictors organized into four construct groups (Table II). This fifteen-variable set is the one used throughout the remainder of the paper — for supervised modelling (Section III-E), SHAP and permutation importance (Sections III-G, III-I), ablation (Section III-K), and unsupervised clustering (Section III-L). Fig. 1 summarizes the audit sequence.

Initial candidate feature set (20 variables)
↓
Leakage audit — near-perfect accuracy flagged as spurious
↓
7 target-derived variables removed
(`account`, `dig_account`, `anydigpayment`, `saved`, `fin22a`, `account_fin`, `account_mob`)
↓
15 leakage-safe predictors (Table II)
↓
Final modelling
(supervised classification, SHAP/permutation/ablation, k-means clustering)

Fig. 1. Leakage audit and correction sequence. The initial specification included target-derived variables and produced spurious near-perfect accuracy; this was identified, corrected, and is reported here for methodological transparency.

TABLE II
FEATURE GROUPS USED IN MODELLING AND ABLATION

Group	Variables
Demographics	`female`, `age`, `educ`, `inc_q`, `in_workforce`, `is_rural`
Access	`con1`, `internet_use`
Behaviour	`merchantpay_dig`, `fin17b`, `fin17c`, `borrowed`, `fin22a_1`
Resilience	`fin24a`, `fin24b`

D. Modelling Objective

Before presenting the supervised pipeline, we state its intended role explicitly. **The objective of supervised modelling in this study was not to develop a deployment-ready individual-level classifier, but to identify and interrogate the behavioural and demographic correlates of adoption** through explainability tooling (SHAP, permutation importance) and diagnostic analysis (calibration, error profiling, ablation). Given a four-level target with substantial class imbalance (68.2% Excluded, 2.1% Basic) and $n = 1{,}000$, a macro F1 in the 0.45–0.50 range is the realistic ceiling for this feature set and sample size, not a shortfall to be corrected by further tuning. We report this ceiling candidly and treat the fitted model as an analytical instrument — a lens for extracting feature-outcome relationships and cross-validating the unsupervised segmentation in Section IV-A — rather than as an endpoint whose accuracy is itself the contribution.

E. Supervised Modelling

Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers were trained as candidate models for extracting adoption correlates, not as competing deployment candidates. All were trained on a stratified 75/25 train-test split, with class weighting applied to address the imbalanced target distribution. The best-performing default configuration (Random Forest) was then re-tuned via `RandomizedSearchCV` (25 iterations, 5-fold stratified cross-validation, scored on macro F1) over a search space covering the number of estimators, maximum depth, minimum samples per leaf, and maximum features considered per split; the full search space is reported in Appendix A. An equivalent search was run for XGBoost for comparison. Tuning aimed to obtain a stable, well-characterized model for explainability analysis, not to maximize headline accuracy.

F. Model Robustness

To assess whether performance estimates were sensitive to the particular train-test split drawn, the tuned Random Forest was additionally evaluated under repeated stratified 5-fold cross-validation with 10 repeats (50 total folds), yielding a distribution of macro F1 scores rather than a single point estimate.

G. Explainability

The tuned Random Forest was explained using SHAP TreeExplainer values. Because the target is multiclass, per-class SHAP arrays were averaged (mean absolute value) to produce an overall importance ranking, and SHAP dependence plots were generated for the top four ranked features against the Advanced-tier class specifically, as this class is the most directly policy-relevant "successful adoption" outcome.

Terminology. Throughout this paper we refer to SHAP, Gini, and permutation rankings as identifying *predictors* or *influential features* — terms describing a variable's contribution to model output — rather than *predictors* or *causes* of adoption. This distinction matters because Findex is a cross-sectional survey (Section III-A) and no causal identification strategy (e.g., instrumental variables, panel data, or experimental variation) is employed here. A feature ranked highly by SHAP is one the model relies on to make predictions; this is informative about association and about which factors are worth investigating further, but it does not establish that changing that feature would change adoption behaviour.

H. Calibration Analysis

Predicted class probabilities were evaluated for calibration using one-vs-rest reliability diagrams (5 quantile-based bins per class) and the Brier score, to assess whether predicted probabilities correspond to observed outcome frequencies – a property distinct from, and not guaranteed by, classification accuracy.

I. Feature Importance Consistency

To assess whether the SHAP-based feature-importance ranking was an artifact of a single explanation method, we additionally computed Gini importance (native to the Random Forest) and permutation importance (20 repeats, scored on macro F1 degradation on the held-out test set), and compared all three rankings via Spearman rank correlation.

J. Error Analysis

Misclassified test-set observations were examined for (i) the dominant confusion pair, (ii) subgroup accuracy by gender and urban/rural status, and (iii) the mean feature profile of misclassified versus correctly classified observations within the dominant confusion pair, to determine whether model errors carried substantive meaning rather than representing unstructured noise.

K. Ablation Study

To independently validate the SHAP-based ranking of construct groups, we conducted a feature-group ablation study, retraining the tuned Random Forest configuration on (i) the full feature set, (ii) each construct group individually held out, and (iii) each construct group used in isolation, measuring macro F1 in each configuration.

L. Unsupervised Segmentation

K-means clustering was applied to the same leakage-safe feature set (standardized via z-score) used in supervised modelling — the fifteen demographic, access, behaviour, and resilience variables in Table II — for $k = 2$ to 5, with k selected by silhouette score and Davies-Bouldin and Calinski-Harabasz indices computed for robustness (Fig. 11).

The Adoption Tier target variable was not included as a clustering input under any configuration. Clustering was performed entirely on demographic, access, behavioural, and resilience predictors; Adoption Tier was used exclusively *after* cluster assignment, to profile and interpret the resulting groups (Table VII). This separation is deliberate: because Adoption Tier is itself constructed from account and digital-usage indicators (Section III-B), including it as a clustering feature would risk the clusters trivially reproducing the target rather than revealing independent structure. Clusters were profiled on demographic and access characteristics and visualized via a two-component PCA projection (Fig. 12); the resulting cluster labels (e.g., "Connected but Unengaged") are therefore *descriptive summaries applied post hoc*, not categories the clustering algorithm was told to find.

a) Defining "Connected but Unengaged.": We use this label to describe a cluster whose members are, on average, *connected* by conventional access metrics (mobile phone ownership, internet use) but *unengaged* by the independently-constructed Adoption Tier (i.e., predominantly in the Excluded tier despite that access). The label names a statistical regularity observed after clustering, not a variable the algorithm optimized for.

M. Cluster Validation

Three checks were used to validate the k-means solution beyond the silhouette score alone. First, bootstrap stability was assessed by refitting k-means on 100 bootstrap resamples and computing the Adjusted Rand Index (ARI) between each resample's cluster assignment (applied to the full original sample) and the original clustering. Second, an algorithm-independent cross-check compared the k-means solution against Ward agglomerative clustering at the same k. Third, because the silhouette score for $k = 3$ was only marginally higher than for $k = 2$ (0.1675), we conducted an explicit interpretability comparison between the two solutions, examining what population-level distinction the additional cluster does or does not preserve. We therefore interpret the three-cluster solution as a substantively useful segmentation of the observed sample rather than evidence that three sharply separated natural populations exist.

N. Statistical Validation

Chi-square tests of independence with Cramer's V effect sizes were used to test associations between cluster membership, Adoption Tier, gender, and urban/rural status. Wilson score 95% confidence intervals were computed for all headline proportions, given the small size of some subgroups (e.g. $n = 21$ for the Basic tier).

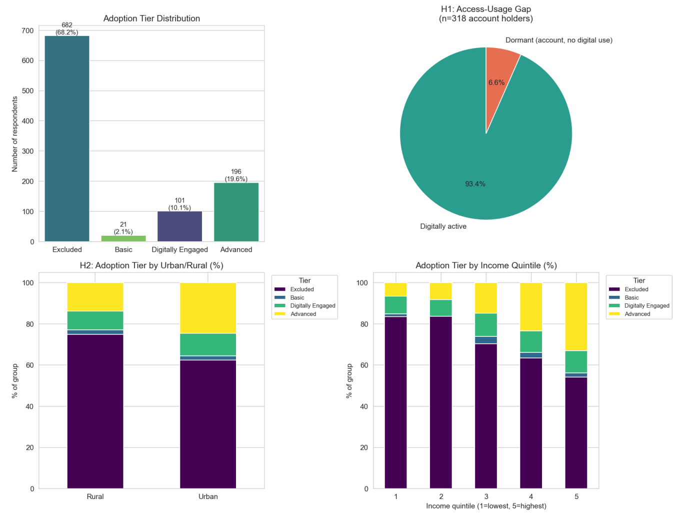

Fig. 2. Adoption Tier distribution, H1 access-usage gap, and Adoption Tier by urban/rural and income quintile.

IV. Results

A. H1 – The Access-Usage Gap (RQ1)

Among the 318 respondents holding a formal account, 6.6% (95% CI: 4.4%–9.9%) showed no digital financial usage – a direct measure of the dormant-account phenomenon described by Chowdhury et al. [1] as a knowledge-behaviour gap in the Bangladesh context.

A substantially larger and more striking pattern emerged from the independent, unsupervised cluster analysis. K-means ($k = 3$), fitted *without access to the Adoption Tier label* (see Section III-L), identified a cluster — labelled "Connected but Unengaged" after the fact based on its profile — comprising 44.3% of the entire sample ($n = 443$). This cluster shows near-universal mobile phone ownership (98.0%) and majority internet use (62.3%), yet 66.8% (95% CI: 62.3%–71.0%) remain formally Excluded from the financial system (Table VII). Because the cluster assignments were derived without using the Adoption Tier labels, the result provides an independent descriptive characterization of a large access-rich, adoption-poor segment, suggesting that digital access alone is insufficient to explain financial participation in this sample.

B. H2 – Urban versus Rural Adoption (RQ1)

Adoption Tier was significantly associated with urban/rural status ($\chi^2 = 21.742$, $p = 0.0001$, $df = 3$, Cramer's V $= 0.147$), and cluster membership showed a parallel association ($\chi^2 = 19.823$, $p < 0.0001$, $df = 2$, Cramer's V $= 0.141$). The direction of the effect favoured urban respondents: rural share declined monotonically across clusters of increasing adoption, from 51.0% in the least-adopted cluster to 46.0% and then 28.6% in progressively more digitally engaged clusters. This is consistent with the conventional urban-adoption advantage reported broadly in the financial inclusion literature, and contrasts with Chowdhury et al.'s reversed finding of a rural literacy advantage in Bangladesh [1].

C. H3 – Predictors of Adoption (RQ3)

TABLE III
SUPERVISED MODEL COMPARISON

Model	Accuracy	F1 (macro)	F1 (weighted)
XGBoost	0.800	0.481	0.781
Random Forest (default)	0.708	0.486	0.737
Random Forest (tuned)	–	**0.494**	–
Decision Tree	0.488	0.424	0.554
Logistic Regression	0.420	0.392	0.473

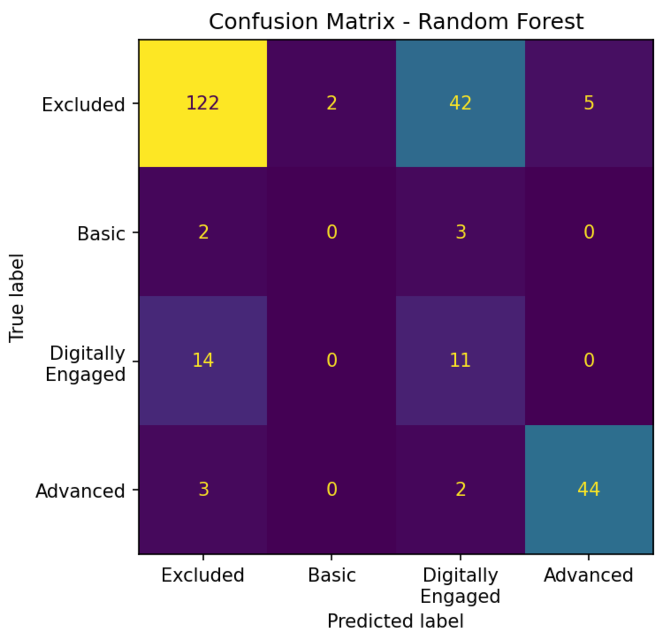

Fig. 3. Confusion matrix for the best-performing model (Random Forest).

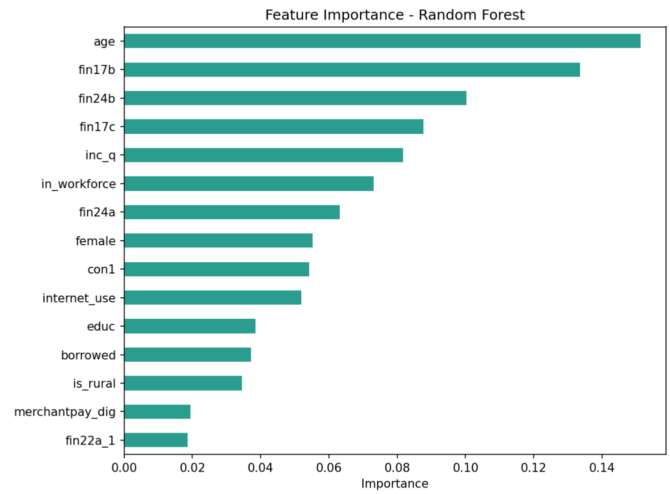

Fig. 4. Random Forest feature importance (Gini-based).

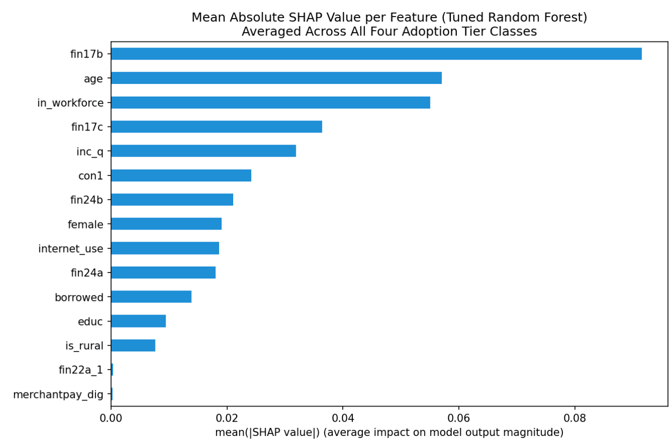

Fig. 5. Mean absolute SHAP value per feature, averaged across all four Adoption Tier classes.

SHAP analysis of the tuned Random Forest (Fig. 5) ranked predictors as follows: saving via mobile money (`fin17b`), age, workforce participation, informal saving (`fin17c`), income quintile, mobile phone ownership, emergency fund coverage time, gender, and internet use. Mobile phone ownership and internet use – the two strongest predictors identified by Saidy and Hassan across Sub-Saharan Africa [3] – ranked in the middle of the importance distribution rather than at the top.

1) SHAP Dependence Plots: Fig. 6 shows dependence plots for the four top-ranked features against the Advanced-tier class. `fin17b` and `fin17c` show clean binary step functions, as expected for binary indicators. Notably, `age` shows a nonlinear relationship: SHAP contribution rises through the twenties and thirties, peaks around ages 40–50, and declines thereafter – indicating that the model's attribution of age to predicted adoption is not simply monotonic, with the largest positive contribution concentrated among economically active, middle-aged respondents.

2) Calibration Analysis: Fig. 7 shows per-class reliability diagrams. The Excluded class is poorly calibrated and systematically underconfident (Brier $= 0.168$): when the model assigns roughly 40% probability to Excluded, the true observed frequency is closer to 65–70%. The Advanced class is comparatively well-calibrated (Brier $= 0.042$), closely tracking the diagonal. The Basic and Digitally Engaged classes show jagged, unreliable calibration curves, consistent with their small effective sample sizes.

3) Feature Importance Consistency: Table IV and Fig. 8 compare Gini, permutation, and SHAP importance rankings. Gini and SHAP rankings agree strongly (Spearman $\rho = 0.914$, $p < 0.0001$); permutation importance agrees more weakly with SHAP ($\rho = 0.504$, $p = 0.056$, not significant at $\alpha = 0.05$) and is negative for several features (`con1`, `female`, `internet_use`, `fin24a`, `educ`, `is_rural`), indicating that shuffling these features does not reliably degrade – and can marginally improve – macro F1 on the test set. `fin17b` is the only feature ranked at or near the top by all three methods, making it the single most robust predictive feature identified in this analysis, subject to the routing-pattern caveat discussed in Section VII.

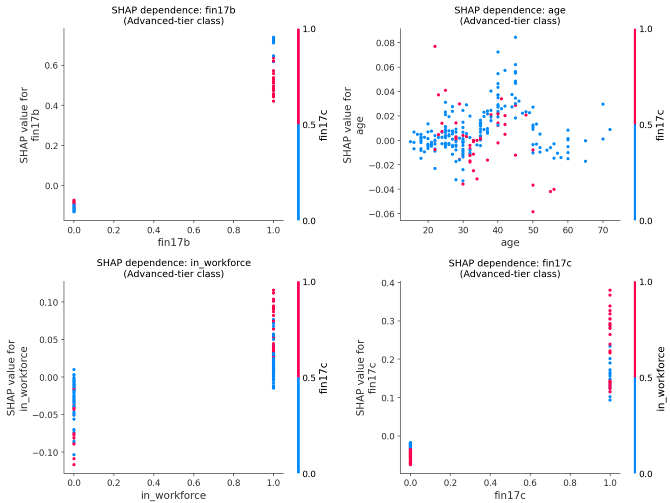

Fig. 6. SHAP dependence plots for the top four features (fin17b, age, in_workforce, fin17c) against the Advanced-tier class, colored by the strongest interacting feature.

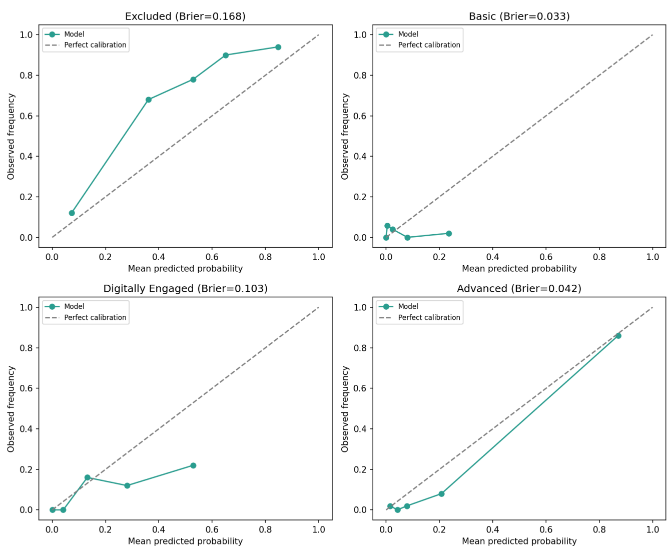

Fig. 7. Per-class reliability diagrams (one-vs-rest) with Brier scores.

Model comparison results (Table III) show XGBoost achieved the highest single-split accuracy (0.800) but Random Forest achieved the highest macro F1 (0.486 default; 0.494 tuned), the metric prioritized here given substantial class imbalance in the target. Consistent with the modelling objective stated in Section III-D, this level of performance is modest in absolute terms and should be read as adequate for extracting directional feature-outcome relationships, not as evidence of a strong individual-level predictive model. All models struggled with the Basic tier ($n = 21$), attributable to sample size rather than model deficiency (Fig. 3); given this small subgroup, per-class metrics for the Basic tier should be treated as indicative only and are not a meaningful test of model quality.

4) Error Analysis: The dominant confusion pair was Excluded predicted as Digitally Engaged (21 cases), with the

TABLE IV
TOP FEATURES BY CROSS-METHOD IMPORTANCE (ABBREVIATED; FULL TABLE IN APPENDIX B)

Feature	Gini rank	Permutation rank	SHAP rank
fin17b	2	1	1
age	1	7	2
in_workforce	5	3	3
fin17c	6	2	4
inc_q	3	6	5

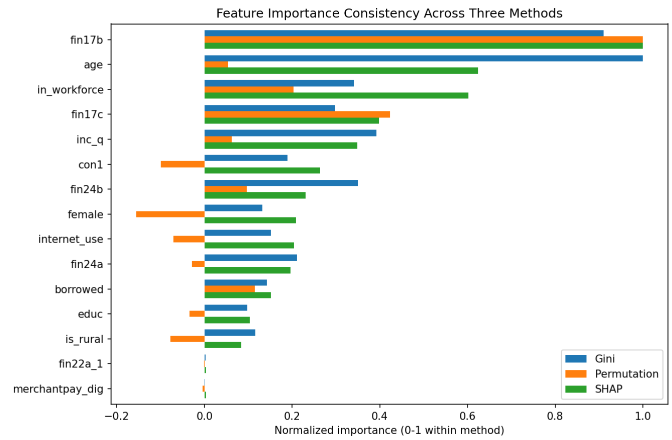

Fig. 8. Normalized feature importance across Gini, permutation, and SHAP methods.

reverse error (Digitally Engaged predicted as Excluded, 16 cases) also common – consistent with this boundary being the hardest to separate given the available features.

Subgroup accuracy was 0.852 for women versus 0.689 for men (ratio 0.808), and 0.780 for rural versus 0.752 for urban respondents (ratio 0.964). We caution against interpreting the gender gap as evidence the model treats women "more fairly": because gender is very strongly associated with cluster membership and outcome in this population (Cramer's V $= 0.807$; see Section IV-E), the majority of women fall into the large, easy-to-predict Excluded class, mechanically inflating aggregate accuracy for that subgroup. We regard this as an important methodological caveat for fairness auditing in financial-inclusion modelling more broadly, discussed further in Section VI.

More substantively, we profiled the feature means of individuals with true label Excluded, comparing those correctly classified against those misclassified as Digitally Engaged (Table V). Misclassified individuals show markedly higher mobile phone ownership (93.5% vs. 49.3%), higher internet use (41.9% vs. 20.7%), higher workforce participation (80.6% vs. 28.6%), and higher income, and are less likely to be female or rural. This profile closely matches the "Connected but Unengaged" cluster identified independently in Section IV-A, providing a second, model-error-based confirmation of the H1 finding through an entirely different analytical route.

5) Ablation Study: Table VI and Fig. 9 show macro F1 under each feature-group configuration. Removing the be-

TABLE V
FEATURE MEANS FOR TRUE-EXCLUDED CASES: CORRECT VS. MISCLASSIFIED

Feature	Misclassified	Correctly Classified
% Female	12.9%	69.3%
Mean age	38.1	33.8
% In workforce	80.6%	28.6%
% Rural	38.7%	51.4%
% Mobile phone (con1)	93.5%	49.3%
% Internet use	41.9%	20.7%
Mean income quintile	3.32	3.13

haviour group causes the largest performance drop ($0.494 \rightarrow 0.415$, -16.1%), while removing the access group slightly *improves* macro F1 ($0.494 \rightarrow 0.506$). Access-only as the sole feature group performs worst of all configurations (macro F1 $= 0.186$, barely above a majority-class baseline). This provides a second, methodologically independent confirmation of the SHAP-based ranking in Section IV-C: behavioural variables carry more predictive weight than raw access variables in this population.

TABLE VI
ABLATION STUDY RESULTS

Configuration	n features	Macro F1
Full feature set	15	0.494
All except demographics	9	0.480
All except access	13	0.506
All except behaviour	10	0.415
All except resilience	13	0.489
Demographics only	6	0.351
Access only	2	0.186
Behaviour only	5	0.389
Resilience only	2	0.305

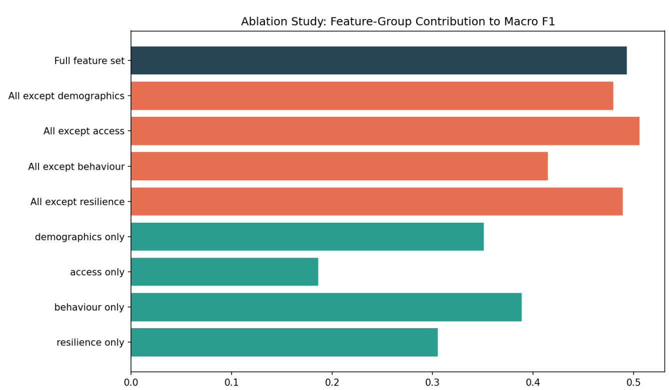

Fig. 9. Ablation study: macro F1 under feature-group leave-one-out and single-group-only configurations.

D. Model Robustness

Repeated stratified cross-validation (50 folds) of the tuned Random Forest yielded a macro F1 distribution with mean $=$ 0.480, std $= 0.023$, and 95% percentile interval $[0.447, 0.521]$ (Fig. 10), tightly bracketing the single-split test estimate of 0.494 and indicating the reported performance is not an artifact of one particular train-test partition.

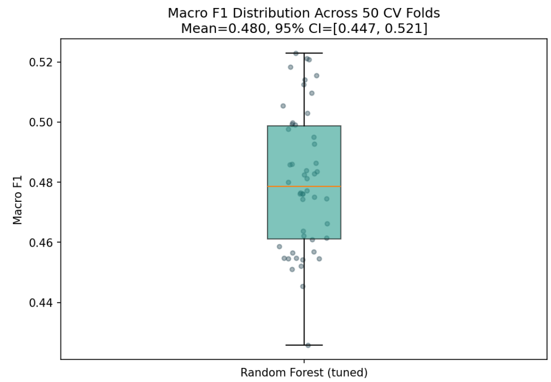

Fig. 10. Distribution of macro F1 across 50 repeated stratified cross-validation folds.

E. Cluster Profiles and Gender Association

This gender effect noted in Section IV-C4 is confirmed with substantially larger magnitude at the cluster level. Cluster membership was very strongly associated with gender ($\chi^2 = 650.549$, $p < 0.0001$, $df = 2$, Cramer's V $= 0.807$), and Adoption Tier itself showed a large association with gender ($\chi^2 = 162.080$, $p < 0.0001$, $df = 3$, Cramer's V $= 0.403$) – both well above the effect sizes observed for urban/rural status. The most digitally excluded cluster was 96.3% female (95% CI: 94.1%–97.7%), compared to 16.3% and 10.3% female in the two more digitally engaged clusters (Table VII).

TABLE VII
CLUSTER PROFILES (K-MEANS, K=3)

Cluster	n (%)	% Female	% Rural	% Mobile	% Excluded
0 – Digitally Excluded Women	431 (43.1%)	96.3%	51.0%	28.3%	89.3%
1 – Connected but Unengaged	443 (44.3%)	16.3%	46.0%	98.0%	66.8%
2 – Advanced Adopters	126 (12.6%)	10.3%	28.6%	98.4%	0.8%

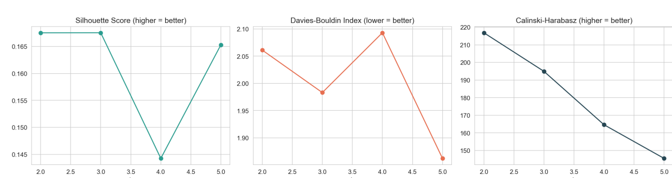

Fig. 11. Silhouette, Davies-Bouldin, and Calinski-Harabasz scores for $k = 2$ to 5.

F. Cluster Validation

Bootstrap resampling (100 replicates) showed the $k = 3$ solution is highly stable: mean ARI $= 0.904$ (std $= 0.041$), 95% percentile interval $[0.847, 0.982]$ (Fig. 14). An algorithm-independent cross-check against Ward agglomerative clustering at the same k showed substantial agreement (ARI

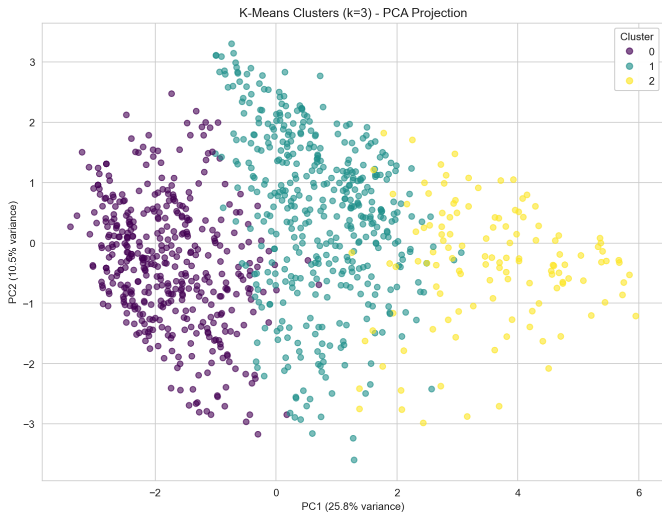

Fig. 12. PCA projection of k-means clusters ($k = 3$).

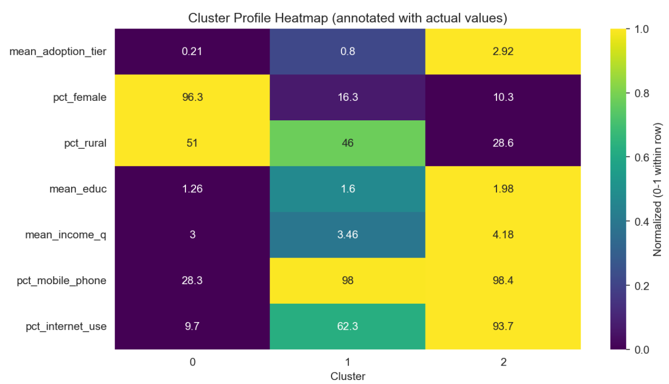

Fig. 13. Normalized cluster profile heatmap across demographic and access variables.

$= 0.740$), indicating the three-cluster structure is not an artifact of the k-means algorithm specifically.

Despite the marginal silhouette difference between $k = 2$ (0.1675) and $k = 3$ (0.1675), the two solutions are substantively different (ARI $= 0.623$). The $k = 2$ solution separates the sample almost entirely along the "Digitally Excluded Women" cluster versus everyone else (515 of 516 members of one $k = 2$ cluster correspond to the original clusters 1 and 2 combined). What $k = 3$ adds is a further split of that "everyone else" group into the access-rich-but-unengaged cluster (66.8% still Excluded) versus the advanced-adopter cluster (0.8% Excluded) — a distinction directly relevant to the access-usage gap central to H1, and the primary substantive justification for reporting $k = 3$ despite its narrow quantitative margin over $k = 2$.

We are explicit about what this evidence does and does not support. The near-identical silhouette scores at $k = 2$ and $k = 3$ indicate that three clusters are not a dramatically better fit to the data's geometry than two; silhouette score alone would not strongly favor either solution. **We therefore interpret $k = 3$**

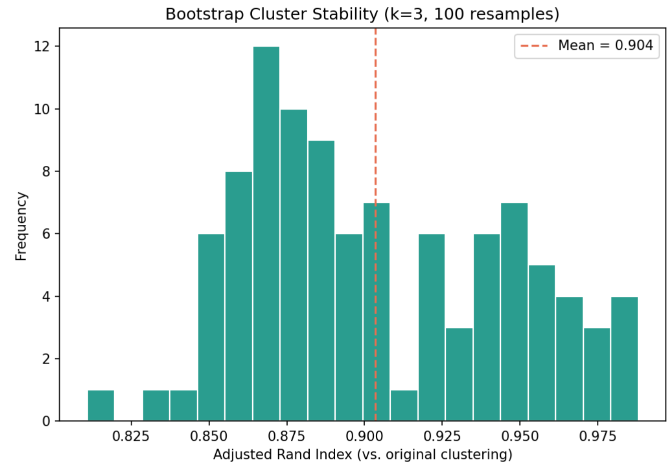

Fig. 14. Distribution of Adjusted Rand Index across 100 bootstrap resamples, evaluating stability of the k=3 clustering solution.

as an analytically useful segmentation — one that recovers a policy-relevant distinction the $k = 2$ solution collapses — rather than as evidence of three sharply separated natural populations. The bootstrap stability (ARI $= 0.904$) and the Ward agglomerative cross-check (ARI $= 0.740$) demonstrate that the $k = 3$ partition is a reproducible way of dividing this sample, not that the underlying population contains three discrete, well-separated subgroups in the geometric sense that silhouette score measures.

V. Discussion

The central contribution of this analysis is the identification of a large, access-rich but adoption-poor population segment, discovered independently through three separate analytical routes: unsupervised clustering (§IV-A), model error analysis (§IV-C4), and feature-group ablation (§IV-C5). These three analyses use different data representations and different statistical logic, yet converge on the same conclusion, summarized in Table VIII.

At 44.3% of the sample, this segment is far larger than the 6.6% dormant-account rate would suggest, and it reframes the practical question facing Pakistan's digital finance sector: for a substantial share of the sample, the observed adoption gap cannot be explained by limited phone or internet access alone, suggesting that downstream factors such as trust, product design, documentation requirements, or social and institutional barriers warrant further investigation. **We are careful to distinguish what this analysis establishes from what it does not: the data identify *who* is Connected but Unengaged and confirm, through three independent methods, that this population is real and substantial — but nothing in a cross-sectional survey of access and demographic variables can identify *why* they remain unengaged.** The candidate mechanisms above are plausible readings of the literature, not findings of this study. The segmentation therefore motivates, rather than resolves, a second-stage mixed-methods investigation into the mechanisms underlying non-adoption

TABLE VIII
THREE INDEPENDENT SIGNATURES OF THE ACCESS-USAGE GAP

Method	What it measures	Result
Unsupervised clustering (§IV-A)	k-means on access/demographic features, blind to Adoption Tier labels	44.3% of sample forms a distinct cluster: 98.0% mobile, 62.3% internet, yet 66.8% formally Excluded
Model error analysis (§IV-C4)	Feature profile of true-Excluded cases the classifier mistakenly predicts as Digitally Engaged	Misclassified group shows far higher mobile ownership (93.5% vs. 49.3%) and internet use (41.9% vs. 20.7%) than correctly classified Excluded cases
Feature-group ablation (§IV-C5)	Macro F1 change when each construct group is removed from the model	Removing *behaviour* features drops macro F1 by 16.1%; removing *access* features *improves* it — access alone is the weakest predictor group (F1 = 0.186)

within this segment — for instance, qualitative interviews or a targeted follow-up survey instrument capable of distinguishing trust deficits from product- design friction from documentation barriers, none of which Findex's existing variables can separate. The convergence of three independent methods on the same population segment substantially strengthens confidence in *who* this population is; it says nothing about *why* they behave this way, and we do not claim otherwise.

A. Practical Implications

The findings above suggest three practical implications for policy and product design in Pakistan's digital-finance ecosystem.

1) Access alone is not enough. Inclusion strategy should separate the pathway from *access* to *adoption* to *active usage*, rather than treating account opening or connectivity as sufficient endpoints.

2) Targeted intervention for the Connected but Unengaged segment. The 44.3% access-rich but adoption-poor population identified here may be better addressed through financial education, trust-building, onboarding assistance, and product redesign than through connectivity expansion alone.

3) Gender-sensitive inclusion design. The strong gender association in both cluster membership and Adoption Tier suggests that generic population-level interventions may miss the most excluded group and should be complemented with gender-responsive targeting and delivery mechanisms.

Interpretive boundary. These implications are hypotheses for intervention, not causal conclusions. They follow from robust associational patterns in this sample and should be tested prospectively through intervention studies or quasi-experimental designs before being treated as policy-certainty claims.

We emphasize that the supervised model's role in this paper is interrogative rather than predictive. A macro F1 of 0.494 (95% CI: [0.447, 0.521] under repeated cross-validation) does not support individual-level deployment, and we do not recommend this classifier, or similarly modest classifiers trained on comparably small survey samples, for operational use in targeting or eligibility decisions (see also Section VI-B). Its value in this study lies instead in the explainability and diagnostic machinery built around it — SHAP rankings, permutation importance, calibration analysis, and error profiling — each of which contributed independent evidence toward the segmentation finding above. We view this study as closer to exploratory behavioural segmentation research than to applied predictive modelling, and believe the paper is stronger, not weaker, for making that distinction explicit.

The gender findings compound this picture. Rather than a marginal demographic control variable, gender shows one of the largest effect sizes in the entire analysis (Cramer's $V = 0.807$ for cluster membership), exceeding even the urban/rural divide by a wide margin, and echoes the gendered impacts of mobile money access documented by Suri and Jack [5] in a causal, longitudinal Kenyan setting. Our cross-sectional Pakistani data cannot establish causality in either direction, but the consistency of a large gender effect across two very different national contexts and methodologies is notable. This diverges from Saidy and Hassan's Sub-Saharan African findings, where account ownership and technology access were the most influential predictors in their models [3], and suggests that regional context is associated with markedly different predictive structures in fintech-adoption models — access-related features dominate in the Sub-Saharan African setting they study, while behavioural and demographic features dominate here. We stress that this is a difference in *which features are predictive*, not a demonstrated difference in *what causes* adoption in each region; we offer it as a hypothesis worth testing with causally-identified designs, not as evidence that access-centric interventions are ineffective in Pakistan. If this predictive pattern does reflect a genuine behavioural difference, it would suggest that access-centric interventions calibrated for Sub-Saharan Africa warrant re-evaluation before being applied to Pakistan, where the access-usage gap and the gender gap appear entangled with each other rather than separate problems.

The H2 finding — urban adoption exceeding rural adoption — aligns with the mainstream financial inclusion literature but diverges from Chowdhury et al.'s Bangladesh result [1]. Since Chowdhury et al. measured financial literacy rather than behavioural adoption, this divergence may reflect a genuine difference between knowledge and behaviour as outcome variables, rather than a contradiction between the two country contexts.

The calibration results carry a practical implication distinct from the accuracy and F1 results typically reported in this literature: a model with modest discriminative performance (macro F1 $\approx$ 0.49) can still be poorly calibrated for its most policy-relevant class. Here, the model systematically underestimates the probability of exclusion, which would understate the scale of the excluded population if predicted probabilities, rather than hard class labels, were used directly for resource targeting. This is a methodological point we believe deserves more attention in applied fintech-adoption modelling generally, where headline accuracy is often reported without accompanying calibration diagnostics.

The feature importance consistency results similarly complicate a simple "top predictor" narrative. While Gini and SHAP rankings agree strongly, permutation importance — arguably the most directly interpretable of the three, since it measures actual performance degradation — agrees only weakly and assigns negative importance to several features, including gender and rural status. We interpret this as evidence that Random Forest's internal (Gini) importance and SHAP's coalition-based attributions can overstate the standalone predictive necessity of a feature when that feature is correlated with others already carrying the relevant signal (e.g., gender's correlation with the cluster structure itself), a distinction between *attribution* and *necessity* that we believe is under-discussed in applied SHAP-based fintech literature. As throughout this paper, we treat all three rankings as statements about model attribution rather than claims about which features causally drive adoption (Section III-G).

VI. Ethical Considerations

A. Fairness and the Accuracy Paradox

Section IV-C4 demonstrated that raw subgroup accuracy (0.852 for women vs. 0.689 for men) is a misleading fairness metric in this setting, because gender is strongly entangled with the outcome variable itself at the population level. A naïve fairness audit relying on aggregate accuracy would conclude the model treats women more favorably, when in fact women are disproportionately concentrated in the largest, easiest-to-predict class. We recommend that fairness audits of financial-inclusion models report subgroup-level macro F1 or per-class recall rather than aggregate accuracy, and we flag this as a specific methodological risk for future work in this domain, since account ownership and gender are correlated by construction in many emerging-market contexts, not just Pakistan.

B. Risk of Misuse

Adoption-tier prediction models of the type developed here are intended to support inclusion-oriented policy and product design – for example, identifying the access-rich, adoption-poor segment as a target for trust-building or product-design interventions. The same predictive infrastructure could, however, be repurposed for exclusionary rather than inclusionary ends: for instance, using predicted low adoption likelihood as an input to credit-scoring or loan-eligibility decisions, which would risk encoding and amplifying existing gender and rural disparities into automated lending decisions. We explicitly do not recommend this model, or models like it, be used for individual-level credit or eligibility decisions; its intended use is aggregate, population-level policy targeting.

C. Human-impact Risk

Because the identified segments correlate strongly with gender and other socioeconomic characteristics, segment labels should not be treated as immutable characteristics of individuals or used to deny services. We recommend treating segment assignment as a provisional, context-dependent signal for outreach design, not as a basis for exclusion, reduced access, or negative automated decision-making at the individual level.

Table IX summarizes the practical scope of this study's modelling approach, separating valid analytical uses from inappropriate individual-level or causal interpretations.

TABLE IX
What the Model Can and Cannot Do

The model can	The model cannot
Identify statistical associations	Establish causality
Characterize population segments	Diagnose individuals
Identify influential model features	Identify causal mechanisms
Support hypothesis generation	Make credit decisions
Motivate targeted research	Determine who "deserves" financial services

D. Calibration and Deployment Caution

Given the poor calibration observed for the Excluded class (Section IV-C2), we caution against using this model's predicted probabilities directly in any downstream resource-allocation formula without recalibration (e.g., Platt scaling or isotonic regression), since the systematic underconfidence identified here would bias such a formula toward underestimating exclusion.

E. Data Privacy

All analysis was conducted on de-identified, publicly available World Bank microdata collected under the Global Findex survey's standard informed-consent protocols. No attempt was made, or would be possible, to re-identify individual respondents from the released microdata.

VII. Limitations

- **Cross-sectional design.** All findings in this paper are associational. Findex is a single-wave cross-sectional survey, and no causal identification strategy — panel data, instrumental variables, or experimental variation — is employed here. No causal claims are made regarding the top predictors of modelled fintech adoption or the mechanisms underlying the gender and access-usage gaps identified; see Section III-G for the predictor/driver terminology convention we follow throughout to keep this distinction explicit at the sentence level, not just in this Limitations section.
- The Basic tier ($n = 21$) is too small for reliable classification by any model tested; this is a sample-size constraint

rather than a modelling failure and should not be overinterpreted. More broadly, with $n = 1{,}000$ split across four imbalanced classes, **this model should not be interpreted as a reliable individual-level classifier** — consistent with the modelling objective stated in Section III-D, its role in this paper is to surface aggregate, population-level correlates of adoption through explainability tooling, not to predict any single respondent's Adoption Tier with confidence. Per-respondent predictions from this model, including the specific misclassified cases discussed in Section IV-C-4, should be read as illustrative of aggregate error patterns, not as reliable statements about those individuals.
- The variable `fin17b` (saved via mobile money), the top-ranked predictor by both Gini and SHAP, is partly confounded with mobile money account ownership due to Findex's survey routing logic (approximately 74% of respondents were routed out of this question); its top ranking should be interpreted with this caveat.
- Permutation importance disagreed substantively with Gini and SHAP for several features, including assigning negative importance to gender and rural status; we interpret this as reflecting correlated-feature attribution effects rather than concluding these features are unimportant, but this disagreement itself is a limitation of relying on any single importance method.
- The three-cluster solution is interpretively and substantively justified (Section IV-F) but is not a dramatically better geometric fit than $k = 2$ by silhouette score alone; see Section IV-F for the explicit distinction we draw between analytical usefulness and natural population separation.
- Calibration is poor for the Excluded, Basic, and Digitally Engaged classes; only the Advanced class is well-calibrated. Predicted probabilities for the other three classes should not be used without recalibration.
- Several chi-square tests involve subgroups with small expected cell counts (notably comparisons involving the Basic tier); results for these specific comparisons should be treated as indicative rather than definitive.
- Subgroup accuracy is confounded by the strong association between gender and outcome in this sample, as discussed in Section VI-A; this is a general risk for fairness auditing in similar settings, not specific to a modelling error here.
- **Generalizability and survey weighting.** All proportions and model estimates reported in this paper — including the 44.3% "Connected but Unengaged" segment size, the 68.2% Excluded rate, and all cluster and subgroup percentages — describe the $n = 1{,}000$ Findex Pakistan sample as analyzed, computed without applying Global Findex's survey design weights. **These figures should not be interpreted as population-representative prevalence estimates for Pakistan as a whole** unless independently reweighted using the Findex sampling weights and design variables, which we did not do here. Findex microdata typically includes weights correcting for unequal selection probability and post-stratification to national demographic totals; because our pipeline treats each respondent as equally weighted, subgroups oversampled or undersampled relative to the true population (e.g., by region, urbanicity, or gender) could shift the reported proportions if corrected. We recommend that any policy application of these findings first replicate the segmentation using appropriately weighted estimates.

VIII. Future Work

Several extensions would strengthen this analysis. First, replicating the pipeline on the Financial Inclusion Insights (FII) Bangladesh tracker survey, if obtained, would provide a genuine same-instrument cross-country comparison. Second, and following directly from the population/mechanism distinction drawn in Section V, a qualitative or mixed-methods follow-up focused specifically on the "Connected but Unengaged" cluster could clarify which downstream barriers — trust, documentation, product design — best explain non-adoption despite access. Third, model recalibration (Platt scaling or isotonic regression) applied specifically to the Excluded class would address the systematic underconfidence identified in Section IV-C2 and should be evaluated before any applied use of predicted probabilities. Fourth, a fairness-aware modelling approach (e.g., equalized-odds post-processing) could be evaluated as a remedy for the subgroup accuracy confound identified in Section VI-A, alongside subgroup-disaggregated macro F1 reporting as a standard diagnostic. Fifth, causal identification strategies (e.g., difference-in-differences around a policy change, or an instrumental-variables approach) would be needed to move beyond the associational claims made throughout this study.

IX. Conclusion

This study suggests that the next frontier of digital financial inclusion in Pakistan may not be connecting people to digital infrastructure, but understanding why access fails to translate into participation. The identification of a large Connected but Unengaged segment provides a measurable target for future mixed-methods and intervention research, while the methodological findings demonstrate the importance of leakage control, calibration analysis, robustness testing, and careful fairness interpretation in financial-inclusion machine learning. Taken together, these findings suggest that the next frontier of financial-inclusion research in Pakistan is not simply expanding digital access, but understanding and addressing the gap between having the means to participate and actually participating.

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Appendix A
Hyperparameter Search Space

TABLE X
HYPERPARAMETER SEARCH SPACE AND SELECTED VALUES

Model	Parameter	Search space (selected)
Random Forest	n_estimators	{200, 300, 500, 800} (300)
	max_depth	{4, 6, 8, 10, None} (None)
	min_samples_leaf	{1, 2, 4, 8} (4)
	max_features	{sqrt, log2, None} (None)
XGBoost	n_estimators	{200, 300, 500} (500)
	max_depth	{3, 4, 5, 6} (6)
	learning_rate	{0.01, 0.03, 0.05, 0.1} (0.1)
	subsample	{0.7, 0.85, 1.0} (0.85)
	colsample_bytree	{0.7, 0.85, 1.0} (0.85)

Table X lists the search space used for `RandomizedSearchCV` (25 iterations, 5-fold stratified CV, scored on macro F1) and the selected best configuration for each model.

Best cross-validated macro F1: Random Forest $= 0.472$; XGBoost $= 0.457$. Held-out test macro F1: default Random Forest $= 0.486$; tuned Random Forest $= 0.494$.

Appendix B
Additional Diagnostic Tables

Table XI reports the complete feature importance comparison across all fifteen features and three methods (Gini, permutation, SHAP), supplementing the abbreviated version in Table IV.

Spearman rank correlations: Gini vs. SHAP $\rho = 0.914$ ($p < 0.0001$); Permutation vs. SHAP $\rho = 0.504$ ($p = 0.0557$); Gini vs. Permutation $\rho = 0.554$ ($p = 0.0323$).

Table XII reports the complete confusion matrix for the tuned Random Forest on the held-out test set ($n = 250$).

Table XIII summarizes the cluster validation metrics reported in Section IV-F.

TABLE XI
FULL FEATURE IMPORTANCE COMPARISON

Feature	Gini	Permutation	SHAP
fin17b	0.2099	0.1087	0.0915
age	0.2307	0.0059	0.0571
in_workforce	0.0786	0.0221	0.0551
fin17c	0.0689	0.0459	0.0364
inc_q	0.0905	0.0067	0.0319
con1	0.0437	−0.0108	0.0242
fin24b	0.0808	0.0105	0.0211
female	0.0306	−0.0169	0.0191
internet_use	0.0349	−0.0077	0.0187
fin24a	0.0486	−0.0031	0.0180
borrowed	0.0329	0.0125	0.0139
educ	0.0225	−0.0038	0.0094
is_rural	0.0266	−0.0085	0.0077
fin22a_1	0.0005	−0.0001	0.0003
merchantpay_dig	0.0004	−0.0005	0.0003

TABLE XII
FULL CONFUSION MATRIX (TUNED RANDOM FOREST, TEST SET)

True \ Predicted	Excluded	Basic	Dig. Engaged	Advanced
Excluded	140	3	21	7
Basic	3	0	2	0
Digitally Engaged	16	0	8	1
Advanced	4	0	2	43

TABLE XIII
CLUSTER VALIDATION SUMMARY

Metric	Value
Bootstrap ARI, mean (std)	0.904 (0.041)
Bootstrap ARI, 95% interval	[0.847, 0.982]
ARI vs. Ward agglomerative	0.740
ARI, k=2 vs. k=3 solution	0.623
Silhouette score, k=2	0.1675
Silhouette score, k=3	0.1675